

High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

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1 Introduction

Quantile treatment effects provide a way to study how the effect of an endogenous treatment varies across different parts of an outcome distribution. This is particularly relevant when the effect at the lower or upper tail differs from the effect near the center. Instrumental variable quantile regression (IVQR) extends structural quantile analysis to settings in which the treatment is endogenous by using excluded instruments together with restrictions on the structural ranks of potential outcomes (Chernozhukov and Hansen 2005). In modern applications, however, researchers may also face a large number of controls, creating an additional high-dimensional nuisance-estimation problem.

Modern applications may also contain many controls, transformations, and interactions. Their coefficients are nuisance parameters, but estimating them without structure can be unstable or infeasible. Regularization makes high-dimensional estimation manageable, while Neyman-orthogonal scores reduce the first-order sensitivity of the target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Chen et al. (2021) combine these ideas with IVQR. Their DML-IVQR procedure accommodates high-dimensional controls and supports candidate-specific inference that does not require strong identification. Orthogonality, however, cannot create identifying information. A valid instrument may contain little information about the endogenous treatment, so moments can distinguish competing treatment-effect values only weakly.

This separation motivates the research gap. Existing DML-IVQR evidence shows how regularized nuisance estimation performs in a high-dimensional design with fixed instrument relevance, but it does not systematically trace how the same procedure behaves as excluded-instrument relevance declines while other structural features remain fixed. Weak-identification-robust inference can retain asymptotic validity without being informative: many false candidates may survive when the instruments provide little treatment variation. Its asymptotic reference distribution can also be imperfectly calibrated in moderate samples. The research question is therefore: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles?

The thesis answers this question through a controlled extension of the Monte Carlo design in Chen et al. (2021). A single parameter, $\kappa \in \{1.00, 0.50, 0.25, 0.10\}$, controls the loading of the excluded-instrument primitives in the treatment equation, with smaller κ corresponding to weaker excluded-instrument relevance within the Monte Carlo design. The extension preserves the marginal distribution of the latent treatment index, the endogeneity channel, the

structural outcome equation, the quantile treatment effect, the controls, and the observed instrument construction. It therefore isolates designed changes in excluded-instrument relevance without treating κ as a universal measure of IVQR identification strength.

The simulations compare Oracle-GMM, an infeasible oracle benchmark; Full-GMM, an unregularized full-control estimator; and DML-IVQR with regularized nuisance estimation on identical samples. Performance is assessed through point-estimation error, the size of numerically accepted sets, power against false candidate values, and coverage at the exact true parameter across several quantiles and two sample sizes.

The evidence gives a consistent answer. As relevance falls, point-estimation errors generally increase, accepted sets become less informative, and power declines. These responses differ across quantiles, with no universal ordering between central and tail quantiles. Coverage does not summarize informativeness and is not uniformly close to its nominal level in finite samples: candidate-specific inference may retain the truth while also retaining many false values, and the asymptotic approximation can under- or over-reject in particular design cells. Regularization often improves on estimating the full high-dimensional nuisance set without regularization, but DML does not dominate in every cell and cannot replace missing instrument information. The central implication is that nuisance robustness and identification robustness address distinct problems; neither guarantees precise or informative inference when excluded instruments contain little information about treatment.

2 Literature Review

Quantile regression describes how covariates are related to different parts of a conditional outcome distribution rather than only its mean (Koenker and Bassett 1978). This distinction matters when treatment effects are heterogeneous: an intervention may shift lower and upper quantiles by different amounts even when its average effect is modest. A structural quantile treatment effect compares the same quantile of potential-outcome distributions under alternative treatment states. It is generally a distributional effect, not an individual-level effect, because the observation occupying a given rank can differ across counterfactual states. Endogeneity creates a separate problem. When realized treatment depends on latent determinants of the outcome, the conditional distribution among observations selecting a treatment value need not equal the potential-outcome distribution obtained by fixing treatment at that value. Ordinary quantile regression of the outcome on treatment and controls then estimates an observed conditional association and does not generally recover a structural quantile effect. The

IVQR framework of Chernozhukov and Hansen (2005) uses excluded instruments to connect endogenous treatment variation to structural outcome quantiles.

Structural IVQR represents potential outcomes through a quantile function and a latent structural rank. For a treatment state d , the structural rank $U_d \in (0, 1)$ is the latent quantile index that locates an individual within the conditional distribution of the potential outcome Y_d given observed controls. Because the structural quantile function is increasing in this rank, a larger value of U_d corresponds to a higher position in the potential-outcome distribution. The rank may be interpreted as summarizing unobserved heterogeneity that determines where an observationally similar individual lies in that distribution. Instrument exogeneity restricts the relationship between the instruments and these structural ranks conditional on controls, while treatment may still depend on latent variables related to the outcome. Rank invariance assumes that an individual's rank is unchanged across treatment states; the weaker rank-similarity condition allows ranks to change but requires their conditional distributions to agree across those states. Under the maintained selection, exogeneity, and rank conditions, the probability that the observed outcome lies below the true structural quantile, conditional on controls and instruments, equals the quantile index (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017). This conditional restriction links the potential-outcome model to estimable instrument moments. Validity permits the instrument to enter those moments; relevance determines whether instrument-induced treatment variation separates competing structural coefficients. Moment validity alone does not establish point identification. A unique solution additionally requires the population mapping from candidate structural quantiles to the restrictions to distinguish the truth from alternatives. That mapping is quantile specific and density weighted, so a mean first-stage statistic is informative about relevance but does not completely characterize IVQR identification at a particular quantile. This distinction is essential empirically: an instrument can satisfy the maintained exogeneity condition yet generate too little treatment variation to locate a structural quantile effect precisely.

When identifying information is strong, the population moment moves sufficiently as the candidate treatment coefficient moves away from its true value, supporting regular point-estimator approximations. With weak information, the true value may remain unique while nearby candidates generate almost indistinguishable moments. Partial identification instead permits more than one population solution, and failed identification leaves some candidates or directions indistinguishable. Chernozhukov and Hansen (2008) develop inference that remains asymptotically valid without imposing the strong-identification conditions needed for conventional estimate-centered inference. Their approach fixes a candidate value, estimates or profiles the nuisance parameters under that candidate, and tests whether the resulting instrument-based

quantile moments are compatible with zero. Inverting these candidate-specific tests retains every non-rejected value and can therefore produce asymmetric, disconnected, or very wide confidence regions. Weak-identification robustness concerns the null reference distribution; it does not guarantee that false candidates will be rejected. Informativeness and finite-sample calibration remain distinct: a robust confidence region can honestly reveal little about the parameter, and an asymptotic test can still reject too often or too rarely in a finite sample. The exact conditional procedure of Chernozhukov et al. (2009) uses a different simulated reference distribution and is not the method implemented here.

The IVQR inference used in this thesis also belongs to a broader literature on inference under weak instrumental-variable identification. Anderson and Rubin (1949) provide the classical foundation for testing structural parameter values through restrictions that remain meaningful without relying on a precisely estimated first stage. Staiger and Stock (1997) formalize weak-instrument asymptotics in linear instrumental-variables models and show why conventional estimate-centered approximations can become unreliable when instruments are only weakly correlated with endogenous regressors. Stock and Wright (2000) extend weak-identification analysis to generalized method-of-moments settings and develop confidence procedures that remain valid when parameters are weakly identified. Alternative robust procedures include the pivotal testing approach of Kleibergen (2002) and the conditional likelihood-ratio method of Moreira (2003). Andrews et al. (2019) provide a modern review of this literature and its implications for applied work. These results motivate the general distinction between conventional estimate-centered inference and candidate-specific procedures that remain valid under weak identification. They do not, however, provide a complete characterization of identification in IVQR. In particular, linear-IV first-stage diagnostics and threshold rules do not directly measure quantile-specific IVQR identification, whose population moments are nonlinear and depend on the distribution of the outcome around the quantile of interest. The thesis therefore uses first-stage statistics only to describe the designed ordering of excluded-instrument relevance, while the IVQR-specific weak-identification-robust inference follows Chernozhukov and Hansen (2008).

High-dimensional controls create a second challenge because estimating the outcome and instrument nuisances may require many coefficients relative to the sample size. Regularization exploits approximate sparsity: a comparatively small set of variables captures most predictive content, while the remaining coefficients are zero or sufficiently small. Shrinkage and selection can nevertheless contaminate a target moment. Double machine learning addresses this problem with a Neyman-orthogonal score whose population expectation has zero first derivative with respect to local nuisance perturbations at the truth (Chernozhukov et al. 2018). Or-

thogonality therefore removes the leading sensitivity to sufficiently small nuisance-estimation errors, not the errors themselves; the learners must still satisfy suitable convergence and regularity conditions. Sample splitting estimates nuisances on one subsample and evaluates the score on another, while cross-fitting rotates these roles across folds. These concepts should not be conflated: orthogonality is a property of the score, whereas cross-fitting organizes data use and limits overfitting dependence. Nor does DML strengthen an instrument. It addresses high-dimensional nuisance estimation, while candidate-specific robust inference addresses irregular inference under limited identification.

Chen et al. (2021) join these two strands. They profile a candidate-specific high-dimensional quantile regression, construct a density-weighted instrument residual, form a Neyman-orthogonal IVQR score, estimate the nuisances with regularization, and evaluate candidates by GMM and weak-identification-robust test inversion. Although they discuss cross-fitting as part of the broader DML framework, the executable DML-IVQR algorithm preserved in this thesis estimates nuisances and evaluates the score on the same sample. Their principal Monte Carlo design holds instrument relevance fixed while comparing Oracle-GMM, Full-GMM, and DML-IVQR across several quantiles. The remaining empirical question is how point-estimation error, confidence-set informativeness, power, and finite-sample calibration respond when excluded-instrument relevance is systematically weakened within that otherwise preserved design. This thesis supplies that controlled stress test; it does not propose a new structural model, score, estimator, identification theorem, or finite-sample correction. Jin and Sun (2025) study a related nonparametric high-dimensional IVQR problem for unconditional quantile treatment effects with automatic DML and cross-fitting, but their target and design differ from the controlled relevance experiment considered here. Chapter 3 states the exact structural model, nuisance construction, estimators, and inference procedure used in the simulations.

3 DML-IVQR Methodology

3.1 Structural IVQR Model and Identification

Following the instrumental-variable quantile regression (IVQR) framework of Chernozhukov and Hansen (2005) and Chernozhukov and Hansen (2008), let Y_i denote a scalar outcome, D_i an endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. The structural quantile function is denoted by $q(u, d, x)$, where

$u \in (0, 1)$ is the structural rank. For a fixed treatment state d , $q(\tau, d, x)$ gives the conditional τ -quantile of the potential outcome obtained by fixing treatment at d .

Under the IVQR assumptions on instrument exogeneity, treatment selection, and rank similarity, the observed data admit the representation

$$Y_i = q(U_i, D_i, X_i), \quad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1), \quad (1)$$

where $U_i := U_{D_i i}$ is the realized structural rank. Rank similarity allows the treatment-state-specific ranks to differ while restricting their conditional distributions; rank invariance is the stronger special case in which the rank is unchanged across treatment states. The strict monotonicity of $q(u, d, x)$ in u links this representation to structural quantiles.

Endogeneity means that the realized treatment D_i may depend on the structural rank U_i . Consequently, the observed conditional quantile of Y_i given D_i and X_i generally differs from the structural quantile obtained by externally fixing treatment. IVQR is designed to recover the latter object. For two treatment states d_1 and d_0 , the structural quantile treatment effect at covariates x is

$$q(\tau, d_1, x) - q(\tau, d_0, x).$$

For a fixed quantile index $\tau \in (0, 1)$, the DML-IVQR framework of Chen et al. (2021) uses a linear-in-parameters structural quantile specification,

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau). \quad (2)$$

The parameter of interest is $\alpha_0(\tau)$, the structural treatment coefficient at quantile τ . The vector $\beta_0(\tau)$ contains the coefficients on the observed controls. The term $c_0(\tau)$ is simply the intercept of the structural quantile equation: it determines the baseline structural quantile when the treatment and nonconstant controls are set to zero. It is written explicitly here rather than being absorbed into the control vector. In the particular Monte Carlo DGP used in Chapter 4, the structural outcome equation has constant term 1, so the true structural intercept is $c_0(\tau) = 1$ for every τ . Keeping $c_0(\tau)$ symbolic in this chapter allows the methodology to be stated in its general form.

The key IVQR restriction follows directly from the structural-rank representation. Because $q(u, d, x)$ is strictly increasing in u ,

$$Y_i \leq q(\tau, D_i, X_i) \iff U_i \leq \tau.$$

Conditional on X_i and Z_i , the structural rank is uniform, so

$$\Pr(U_i \leq \tau \mid X_i, Z_i) = \tau.$$

Substituting the linear structural quantile specification therefore gives

$$\Pr[Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau) \mid X_i, Z_i] = \tau. \quad (3)$$

Throughout the remainder of the thesis, $\mathbf{1}\{A\}$ denotes the indicator function: it equals one when statement A is true and zero otherwise. The probability restriction can therefore be written as the conditional moment

$$\mathbb{E}[\tau - \mathbf{1}\{Y_i \leq c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau)\} \mid X_i, Z_i] = 0. \quad (4)$$

This equation is the central population restriction used by IVQR. At the true structural parameters, the indicator has conditional expectation τ . A candidate treatment coefficient that is inconsistent with the structural quantile restriction will generally produce a nonzero moment. Instrument validity makes these moments legitimate; instrument relevance determines how strongly they change when the candidate treatment coefficient moves away from the truth. Hence validity and identification are distinct. A valid instrument can still contain too little treatment variation to distinguish competing values of $\alpha_0(\tau)$ precisely. The next subsection shows how DML-IVQR constructs a nuisance-robust version of this candidate-specific moment.

3.2 Orthogonal Score and Nuisance Estimation

Fix a quantile τ and consider a candidate value a for the structural treatment coefficient $\alpha_0(\tau)$. IVQR evaluates such candidate values one at a time. For a given a , subtracting $D_i' a$ from the outcome leaves an adjusted outcome whose conditional τ -quantile must be explained by an intercept and the controls. The first nuisance-estimation problem therefore profiles these quantities conditional on the candidate a .

Define the augmented control vector

$$\widetilde{X}_i = (1, X_i')',$$

where the first component is the constant and the remaining components are the p nonconstant controls. Quantile regression measures fit with the check-loss function

$$\rho_\tau(u) = u[\tau - \mathbf{1}\{u < 0\}]. \quad (5)$$

Here u is a generic scalar residual argument of the check-loss function. The check loss is asymmetric: observations above and below the candidate quantile receive weights determined by τ . Minimizing this loss therefore fits the conditional τ -quantile rather than the conditional mean.

Following Chen et al. (2021), for each candidate a , let $c(a)$ denote the profiled intercept and let $\beta(a)$ denote the profiled vector of control slopes. They are defined jointly by

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg \min_{c,b} \mathbb{E}[\rho_\tau(Y_i - D_i' a - c - X_i' b)]. \quad (6)$$

Here $c \in \mathbb{R}$ is the optimization variable for the intercept and $b \in \mathbb{R}^p$ is the optimization vector for the control slopes. Their minimizers are $c(a)$ and $\beta(a)$. These are candidate-specific nuisance parameters: changing a changes the adjusted outcome $Y_i - D_i' a$, and therefore generally changes the fitted intercept and control coefficients. At the true candidate $a = \alpha_0(\tau)$, maintained uniqueness gives $c(a) = c_0(\tau)$ and $\beta(a) = \beta_0(\tau)$.

$$\mathbb{E}[\{\tau - \mathbf{1}[Y_i \leq D_i' a + c(a) + X_i' \beta(a)]\} \tilde{X}_i] = 0. \quad (7)$$

The moment above is the first-order quantile-regression condition for the profiled intercept and controls. It says that, at the fitted candidate-specific quantile, the quantile residual is orthogonal to the augmented controls.

With the candidate treatment effect and profiled nuisance coefficients fixed, define the candidate residual

$$\epsilon_i(a) = Y_i - D_i' a - c(a) - X_i' \beta(a). \quad (8)$$

The residual equals zero at the candidate fitted quantile. The local behavior of a quantile moment depends on how much probability mass lies near this zero boundary. This is why the residual density at zero enters the orthogonalization. Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ be its conditional density at zero.

Suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\tilde{X}_i \in \mathbb{R}^{p+1}$. The next step removes from the instruments the component that is locally associated with the controls. Because quantile moments change according to the residual density at zero, this residualization is density weighted rather than an ordinary unweighted linear projection. Define

$$\begin{aligned} M(a) &= \mathbb{E}[Z_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathbb{E}[\tilde{X}_i \tilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)], \quad J(a) \in \mathbb{R}^{(p+1) \times (p+1)}. \end{aligned} \quad (9)$$

Assuming $J(a)$ is nonsingular, set

$$\delta(a) = M(a)J(a)^{-1}, \quad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \quad (10)$$

Thus $J(a)$ describes density-weighted variation in the augmented controls, while $M(a)$ describes their density-weighted relationship with the instruments. The matrix $\delta(a) = M(a)J(a)^{-1}$ contains the coefficients of the corresponding population residualization. Partition this matrix as $\delta(a) = [d_0(a), \Gamma(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\Gamma(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\tilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \quad (11)$$

The vector $\Psi_i(a)$ is therefore the instrument after removing the density-weighted component associated with the intercept and controls. It is this residualized instrument, rather than the raw Z_i , that enters the theoretical orthogonal score. By construction, $E[(Z_i - \delta(a)\tilde{X}_i)\tilde{X}_i' f_{\epsilon(a)}(0 | X_i, Z_i)] = 0$; this is not an ordinary unweighted residual or a conventional first-stage coefficient.

The candidate-specific nuisance vector collects the quantities required to construct the score.

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \text{vec}\{\delta(a)\} \end{pmatrix}, \quad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \text{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \quad (12)$$

Vectorization only stacks the residualization matrix. The target remains $\alpha_0(\tau)$; $c(a)$ and $\beta(a)$ profile the outcome nuisance, while $\delta(a)$ residualizes the instruments.

Following Chen et al. (2021), the orthogonal score combines two objects: the candidate quantile residual and the density-weighted residualized instrument.

$$\begin{aligned} g_\tau(W_i; a, \eta) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c + X_i' \beta\}] (Z_i - \delta \tilde{X}_i), \\ g_\tau(W_i; a, \eta(a)) &= [\tau - \mathbf{1}\{Y_i \leq D_i' a + c(a) + X_i' \beta(a)\}] \Psi_i(a), \end{aligned} \quad (13)$$

where $W_i = (Y_i, D_i, X_i, Z_i)$. The first factor asks whether the observed outcome lies below the candidate structural quantile. The second factor supplies instrument variation after removing the control-related component. At the true treatment coefficient and true nuisance values, their population product has expectation zero.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$E[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \quad (14)$$

Ordinary plug-in estimation of high-dimensional nuisances can contaminate a target moment. DML therefore uses a score with the stronger local property that its population expectation is

insensitive, to first order, to small perturbations of the nuisance vector at the truth. Formally,

$$\partial_{\eta} E[g_{\tau}(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = 0. \quad (15)$$

Here ∂_{η} denotes differentiation with respect to the nuisance vector. Orthogonality does not mean that nuisance estimation is error free. It means that sufficiently small nuisance-estimation errors do not enter the target moment at first order.

The cancellation can be seen directly. First consider perturbations of the profiled intercept and control slopes. To show the cancellation for the intercept and slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$ and define

$$\begin{aligned} M_0 &= E[Z_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ J_0 &= E[\tilde{X}_i \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \quad (16)$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned} \partial_{(c, \beta')} E[g_{\tau}(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} &= -E[(Z_i - \delta_0 \tilde{X}_i) \tilde{X}_i' f_{\epsilon_0}(0 | X_i, Z_i)] \\ &= -[M_0 - \delta_0 J_0] = 0. \end{aligned} \quad (17)$$

The density appears because observations near the boundary can switch sides. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\tilde{X}_i$, cancelling first-order changes in the intercept and slopes.

A similar argument applies to the instrument-residualization nuisance. For instrument component $j = 1, \dots, d_z$, let δ_j denote the corresponding row of δ . Then

$$\partial_{\delta_j} E[g_{\tau, j}(W_i; \alpha_0(\tau), \eta)]|_{\eta=\eta_0} = -E[\{\tau - \mathbf{1}[Y_i \leq D_i' \alpha_0(\tau) + c_0(\tau) + X_i' \beta_0(\tau)]\} \tilde{X}_i'] = 0. \quad (18)$$

An error in δ is multiplied by the quantile residual, whose moment with $\tilde{X}_i$ is zero by (7), so its first-order effect also vanishes. This construction permits sufficiently accurate regularized nuisances to support inference on the low-dimensional target (Chernozhukov et al. 2018; Chen et al. 2021).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a . IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.3 Estimators and Executable Implementation

Throughout this methodology section, N denotes the sample size. Chapter 4 uses n for the corresponding Monte Carlo design sample size. Section 3.2 defined population nuisances. When controls are numerous relative to N , DML-IVQR estimates their sample counterparts with ℓ_1 regularization under approximate sparsity: a relatively small group has substantial predictive content while other coefficients are zero or small.

Fix τ and a candidate value a . Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D'_i a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \dots, b_p)'$ contain only the slopes on X_i . Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$. The implementation-specific notation for the later instrument regression is defined below and is kept distinct from the full population matrix $\delta(a) = [d_0(a), \Gamma(a)]$. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \arg \min_{b_0, b} \left\{ \sum_{i=1}^N \rho_\tau(Y_i - D'_i a - b_0 - X'_i b) + \lambda_0(a)|b_0| + \sum_{k=1}^p \lambda_k(a)|b_k| \right\}. \quad (19)$$

The first term measures fit at quantile τ . The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum because this is the scaling used by the fitted quantile-regression routine. Following the pivotal route described by Chen et al. (2021), let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \dots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The implemented penalty is

$$\begin{aligned} T_r &= \max_{1 \leq k \leq p} \left| \frac{\sum_{i=1}^N X_{ik} [\tau - \mathbf{1}\{U_{ir} < \tau\}]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda(a) &= 2 \widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000}) \sqrt{\tau(1-\tau)}(1, s_1, \dots, s_p)'. \end{aligned} \quad (20)$$

Here $k = 1, \dots, p$ indexes the control variables, $r = 1, \dots, 1000$ indexes the simulated pivotal draws, T_r is the simulated pivotal statistic in draw r , and $\widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000})$ denotes their empirical 90th percentile. The $p + 1$ entries of $\lambda(a)$ correspond respectively to the intercept and the p control slopes. Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on

the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned}\widehat{\epsilon}_i(a) &= Y_i - D_i' a - \widehat{b}_0(a) - X_i' \widehat{\beta}(a), \\ \widehat{f}_i(a) &= \phi\left(\widehat{\epsilon}_i(a); \bar{\epsilon}(a), s_{\bar{\epsilon}(a)}^2\right),\end{aligned}\tag{21}$$

Here $\bar{\epsilon}(a)$ denotes the sample mean of the N candidate residuals and $s_{\bar{\epsilon}(a)}^2$ denotes their sample variance. where $\phi(e; \mu, \sigma)$ is the normal density with mean μ and standard deviation σ . Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate residuals.¹ This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 | X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component $j = 1, \dots, d_z$, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{r}_{j0}(a)$ denote its retained ordinary intercept and let $\widehat{\Gamma}_{\text{imp},j}(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix} \widehat{r}_{j0}(a) \\ \widehat{\Gamma}_{\text{imp},j}(a) \end{pmatrix} = \arg \min_{(r_{j0}, \gamma_j)} \left\{ \sum_{i=1}^N \left[\sqrt{\widehat{f}_i(a)} Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)} X_i' \gamma_j \right]^2 + \mathcal{P}_{j,a}(\gamma_j) \right\}.\tag{22}$$

Here $\mathcal{P}_{j,a}(\gamma_j) = \sum_{k=1}^p \lambda_{jk}^F(a) |\gamma_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. Here $\gamma_j \in \mathbb{R}^p$ is the slope vector for instrument component j , and $k = 1, \dots, p$ indexes its control coefficients. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the intercept and the slopes. Define $\widehat{r}_0(a) = (\widehat{r}_{10}(a), \dots, \widehat{r}_{d_z 0}(a))'$ and stack the slope rows in $\widehat{\Gamma}_{\text{imp}}(a)$. The residualized instrument used in the score is

$$\begin{aligned}\widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i' \widehat{\Gamma}_{\text{imp},j}(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{\Gamma}_{\text{imp}}(a) X_i.\end{aligned}\tag{23}$$

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide the retained intercept by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The intercept retained by the frozen transformed regression is implementation specific. It enters as an ordinary constant rather than $\sqrt{\widehat{f}_i(a)}$ times a constant regressor, so $\widehat{r}_0(a)$ is not the literal sample counterpart of $d_0(a)$ in the augmented population projection of Section 3.2. The exact derivative cancellation in Section 3.2 applies to the theoretical augmented score. The executable score follows its density-weighted slope residualization but treats the intercept differently. The results therefore characterize the preserved implementation, not a literal sample implementation of every component of the theoretical score.

The penalties can shrink weak coefficients to zero and stabilize both high-dimensional steps, but also create shrinkage error. The theoretical score in Section 3.2 limits its first-order effect (Chernozhukov et al. 2018). Nuisances and scores use the same sample; this preserved implementation has no sample splitting or cross-fitting.

Oracle-GMM, Full-GMM, and DML-IVQR target the same $\alpha_0(\tau)$ using the same outcome, treatment, instruments, and quantile. Each multiplies a candidate quantile score by residualized instruments, producing a moment with dimension d_z ; nuisance estimation differs.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \dots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\text{orc}}(a) \\ \widehat{\beta}_{\text{orc}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_{S_0,i}' b). \quad (24)$$

Thus, Oracle-GMM first removes $D_i' a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\text{full}}(a) \\ \widehat{\beta}_{\text{full}}(a) \end{pmatrix} = \arg \min_{b_0, b} \sum_{i=1}^N \rho_\tau(Y_i - D_i' a - b_0 - X_i' b). \quad (25)$$

Thus all 100 controls enter freely. Full-GMM does not require the unknown oracle set, but estimating many weak or irrelevant coefficients can be noisy.

Oracle-GMM and Full-GMM use the same unregularized instrument residualization with their respective control matrices. For $e \in \{\text{orc}, \text{full}\}$, let $X_{\text{orc},i} = X_{S_0,i}$ and $X_{\text{full},i} = X_i$, and let $\widehat{f}_{e,i}(a)$ be the fitted-normal proxy defined above. The code computes

$$\begin{aligned}\widehat{M}_e(a) &= \frac{1}{N} \sum_{i=1}^N Z_i X'_{e,i} \widehat{f}_{e,i}(a), & \widehat{J}_e(a) &= \frac{1}{N} \sum_{i=1}^N X_{e,i} X'_{e,i} \widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a) \widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}.\end{aligned}\tag{26}$$

This direct, unpenalized inversion includes no residualization intercept, although the quantile regression has one, and uses the original Z_i in the final residual. It therefore differs from the augmented population orthogonalization in Section 3.2. There is no ridge adjustment or generalized inverse: a singular $\widehat{J}_e(a)$ produces a recorded numerical failure. Each benchmark multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR uses all controls and regularizes both nuisance steps. It estimates the control coefficients by quantile Lasso, forms the density proxy, and uses regularized weighted instrument regressions. DML-IVQR-BC applies (20). Oracle and Full also use residualized IVQR moments, but estimate nuisances without ℓ_1 regularization. Unlike their matrix calculation, DML retains a residualization intercept. The comparison covers a known oracle set without regularization, all controls without regularization, and all controls with regularized nuisance estimation. It does not isolate every implementation difference but keeps the structural model, target, and instruments common.

3.4 GMM Criterion and Weak-Identification-Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and $\widehat{g}_{e,i}(a) \in \mathbb{R}^{d_z}$ its observation-level score. Its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a).\tag{27}$$

This vector measures the sample restriction at a . The code measures its variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N} \sum_{i=1}^N \widehat{g}_{e,i}(a) \widehat{g}_{e,i}(a)'. \tag{28}$$

This matrix standardizes differently scaled and correlated components. It uses denominator N , no finite-sample correction, and no sample-mean centering. At the truth, its probability limit equals the score covariance because the population moment is zero.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N\widehat{g}_{e,N}(a)' \widehat{\Sigma}_e(a)^{-1} \widehat{g}_{e,N}(a). \quad (29)$$

This equals the code's calculation from the unnormalized score sum and sum of outer products. Smaller values indicate moments closer to zero after standardization. The direct inverse has no ridge or generalized inverse; failure or a nonfinite result is recorded. Suppressing e gives $W_N(a)$. Let $\mathcal{A}$ denote the set of candidate treatment-effect values over which the GMM profile is evaluated; its finite numerical specification in the Monte Carlo experiment is given in Section 4.3. The point estimator is

$$\widehat{\alpha}_e(\tau) = \arg \min_{a \in \mathcal{A}} W_{e,N}(a). \quad (30)$$

The minimum need not be zero. The implementation minimizes over the finite profile in Section 4.3, selects the first exact tie, and gives no point estimate to an incomplete profile.

Point estimation retains the minimizer; inference tests each a separately. Under the regularity conditions for the corresponding theoretical candidate-specific IVQR score, the null statistic has the asymptotic reference distribution

$$W_{e,N}(a) \xrightarrow{d} \chi_{d_z}^2, \quad d_z = \dim(Z_i). \quad (31)$$

The degrees of freedom equal the number of moment components. The preserved Monte Carlo implementation applies this chi-square law as the asymptotic reference distribution for the candidate-specific statistics of all three estimators. For DML-IVQR-BC, however, the implementation-specific residualization intercept described in Section 3.3 is not the literal sample counterpart of the augmented theoretical intercept. This thesis therefore does not establish a separate limit theorem for that implementation detail. Following Chernozhukov and Hansen (2008), the test evaluates the hypothesized value directly rather than using a normal approximation for $\widehat{\alpha}_e(\tau)$. At level γ , reject when $W_{e,N}(a) > c_{1-\gamma, d_z}$. Here $\gamma \in (0, 1)$ is the test significance level, and $c_{1-\gamma, d_z}$ is the $(1 - \gamma)$ -quantile of a chi-square distribution with d_z degrees of freedom. This test-level γ is distinct from the instrument-regression slope vector γ_j used in Section 3.3.

Inverting these candidate-specific tests gives

$$C_{e,1-\gamma}(\tau) = \{a \in \mathcal{A} : W_{e,N}(a) \leq c_{1-\gamma, d_z}\}. \quad (32)$$

Test inversion retains every candidate not rejected by its own moment test. Because the profile can cross the cutoff repeatedly, the region may be connected, disconnected, wide, or empty. A finite numerical domain describes only acceptance within that domain and cannot establish global unboundedness.

The underlying Chernozhukov–Hansen test-inversion approach can retain asymptotic validity under weak, partial, or failed identification, subject to its regularity conditions, but need not be informative. Weak instruments can leave many candidates below the cutoff. Because the implementation uses an asymptotic chi-square critical value, finite-sample calibration remains empirical. This differs from the exact conditional simulation method of Chernozhukov et al. (2009).

4 Monte Carlo Design

Chapter 3 uses N as generic sample-size notation; the simulation chapters use n for the design sample size.

4.1 Data-Generating Process and Instrument-Relevance Extension

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i , the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \quad (33)$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (33), and they satisfy

$$x_{ji} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1), \quad j = 1, \dots, 100, \quad z_{1i}, z_{2i}, v_{1i}, v_{2i} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1). \quad (34)$$

Observations are independent across i . The observed controls are

$$X_{ji} = \Phi(x_{ji}), \quad j = 1, \dots, 100, \quad (35)$$

The observed control vector is $X_i = (X_{1i}, \dots, X_{100i})'$. Here Φ is the standard-normal cumulative distribution function, so $X_{ji} \sim \mathcal{U}(0, 1)$ independently across controls.

The executable design constructs two instruments as²

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \quad (36)$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \quad (37)$$

Thus $Z_i = (Z_{1i}, Z_{2i})'$, so the number of instrument components is $d_z = 2$. The z components enter the treatment index directly, the included controls induce dependence between instruments and controls, and the v components add independent noise. The executable baseline treatment is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \quad (38)$$

The treatment is continuous and lies in $(0, 1)$.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5 \sum_{j=1}^7 X_{ji} + D_i u_i. \quad (39)$$

The outcome contains $X_{1i}, \dots, X_{7i}$; together with the controls in the instruments, $X_{1i}, \dots, X_{10i}$ form the ten-control oracle set. For $d \in (0, 1)$, Equation (39) implies

$$Y_i(d) = 1 + 5 \sum_{j=1}^7 X_{ji} + d(1 + u_i). \quad (40)$$

Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^7 X_{ji} + d [1 + \Phi^{-1}(\tau)]. \quad (41)$$

Here Φ^{-1} denotes the standard-normal quantile function, i.e. the inverse of the standard-normal cumulative distribution function Φ . The structural quantile treatment effect is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \quad (42)$$

In particular, $\alpha_0(0.50) = 1$.

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j . The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

The extension varies excluded-instrument relevance through $\kappa \in [0, 1]$. Let $w_i \sim \mathcal{N}(0, 1)$ be independent of all preceding primitives. The latent index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)} w_i + \epsilon_i, \quad (43)$$

$$D_i(\kappa) = \Phi(d_i^*(\kappa)). \quad (44)$$

This symmetric loading is the sole structural modification to the executable baseline.

At $\kappa = 1$, the coefficient on w_i is zero and

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \quad (45)$$

which reproduces Equation (38). Independence and unit variances imply

$$\text{Var}[d_i^*(\kappa)] = 2\kappa^2 + 2(1 - \kappa^2) + 1 = 3. \quad (46)$$

Thus $d_i^*(\kappa) \sim \mathcal{N}(0, 3)$ for every κ , preserving the marginal distribution of $D_i(\kappa)$. This distribution is not uniform because Φ is applied to an index with variance three.

Instrument relevance changes because

$$\text{Cov}(Z_{1i}, d_i^*(\kappa)) = \text{Cov}(Z_{2i}, d_i^*(\kappa)) = \kappa. \quad (47)$$

All other instrument components are independent of $d_i^*(\kappa)$, so lower κ reduces each instrument–index covariance. The parameter governs relevance but is not a complete measure of IVQR identification. Endogeneity remains fixed because u_i is independent of z_{1i} , z_{2i} , and w_i while its covariance with ϵ_i is unchanged:

$$\text{Cov}(u_i, d_i^*(\kappa)) = \text{Cov}(u_i, \epsilon_i) = 0.3. \quad (48)$$

Joint normality and the invariant variance and covariance in Equations (46) and (48) keep the joint distribution of $(u_i, d_i^*(\kappa))$ unchanged. The controls, observed-instrument construction, structural outcome equation, induced marginal outcome distribution, and structural quantile effect are also unchanged. The extension therefore isolates relevance from the marginal treatment distribution, endogeneity channel, and structural treatment-effect distribution.

4.2 Simulation Design and Estimator Comparison

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} n &\in \{500, 1000\}, & \tau &\in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ \kappa &\in \{1, 0.50, 0.25, 0.10\}, & R &= 500. \end{aligned} \quad (49)$$

Here n is the Monte Carlo sample size, τ is the quantile index, κ is the instrument-relevance design parameter, and $R = 500$ is the number of Monte Carlo replications in each (n, τ, κ) design. The Cartesian product contains 40 distinct (n, τ, κ) designs, each with 500 replications.

Oracle-GMM uses the known oracle control set; Full-GMM uses all controls without regularization; and DML-IVQR uses regularized high-dimensional nuisances and an orthogonalization-based score construction, with the implementation-specific residualization intercept described in Section 3.3. The design therefore has 120 design–estimator cells. Every comparison uses the truth $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$ from Equation (42).

All estimators use the same sample within a replication. DML-IVQR uses the BC pivotal penalty and same-sample procedure in Section 3.3. Inference is conducted at 95%, and power uses the alternatives defined below.

4.3 Performance Measures and Numerical Inference

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r , and let $\alpha_0(\tau)$ be the true value from Equation (42). Replications are indexed by $r = 1, \dots, R$. Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^R [\alpha_0(\tau) - \widehat{\alpha}_r(\tau)], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^R |\widehat{\alpha}_r(\tau) - \alpha_0(\tau)|, \\ \text{RMSE}(\tau) &= \left\{ \frac{1}{R} \sum_{r=1}^R [\widehat{\alpha}_r(\tau) - \alpha_0(\tau)]^2 \right\}^{1/2}. \end{aligned} \quad (50)$$

Measures are computed separately by estimator and design cell. Positive Bias means estimates lie below the truth on average; MAE records absolute error, and RMSE gives greater weight to large errors.

When the generic statistic W_N from Chapter 3 is evaluated in the Monte Carlo experiment, N equals the design sample size n ; the additional subscript r identifies the Monte Carlo replication. With $c_{0.95}$ denoting the critical value defined below, define

$$C_r(\tau) = \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) \leq c_{0.95}\}, \quad \widehat{\text{Coverage}}(\tau) = \frac{1}{R} \sum_{r=1}^R C_r(\tau). \quad (51)$$

Coverage evaluates $W_{N,r}$ directly at $\alpha_0(\tau)$ and is therefore independent of the numerical grid. Empirical size is

$$\widehat{\text{Size}}(\tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\} = 1 - \widehat{\text{Coverage}}(\tau). \quad (52)$$

The equality follows from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta, \quad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \quad (53)$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{W_{N,r}(a_{\Delta}(\tau)) > c_{0.95}\}. \quad (54)$$

The statistic is evaluated directly at each false value, independently of the confidence-region grid; $\Delta = 0$ is size, not power.

The accepted-set measure totals all accepted components inside the primary computational domain, including disconnected components. It is neither the span between extreme accepted values nor an unrestricted confidence-region length. Because $d_z = 2$, the 95% cutoff for the statistic in Section 3.4 is

$$c_{0.95} = \chi_{2,0.95}^2 = 5.991464547 \dots \quad (55)$$

A candidate is accepted when $W_N(a) \leq c_{0.95}$. This asymptotic cutoff is not an exact finite-sample guarantee. The primary grid is

$$\begin{aligned} A_0 &= [-1, 3], & h &= 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \dots, 41\}, \\ &= \{-1, -0.9, \dots, 2.9, 3\}. \end{aligned} \quad (56)$$

Here $A_0 = [-1, 3]$ is the fixed primary computational domain, $h = 0.1$ is the distance between adjacent grid points, j is the grid-point index, and a_j denotes the j -th candidate treatment-effect value. The set G_0 contains all 41 candidate values. There are 41 points because the

four-unit interval from -1 to 3 , evaluated in increments of 0.1 and including both endpoints, contains $4/0.1 + 1 = 41$ grid points. The same grid is used throughout. A_0 is a computational domain, not an asserted parameter space, so reaching its boundary establishes neither global boundedness nor unboundedness.

The initial numerical design considered reporting total confidence-region length after adaptive domain expansion. Numerical grid diagnostics showed that, under weak relevance, accepted sets could remain non-localized over increasingly wide numerical domains. The final design therefore reports a fixed-domain accepted-set measure and treats domain expansion as a separate numerical diagnostic rather than assigning an artificial finite confidence-region length.

For replication r , define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\{W_{N,r}(a_j) \leq c_{0.95}\}, \\ C_{r,G_0} &= \{a_j \in G_0 : I_r(a_j) = 1\}, \\ F_r &= \mathbf{1}\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\}. \end{aligned} \quad (57)$$

For the primary domain A_0 , define left-boundary, right-boundary, and either-boundary acceptance in replication r by

$$B_r^L = I_r(a_1), \quad B_r^R = I_r(a_{41}), \quad B_r^E = \mathbf{1}\{B_r^L = 1 \text{ or } B_r^R = 1\}. \quad (58)$$

The corresponding either-boundary acceptance rate over R Monte Carlo replications is

$$\widehat{B}^E = \frac{1}{R} \sum_{r=1}^R B_r^E. \quad (59)$$

The same definition is used for an expanded diagnostic domain, with its own left and right endpoints. A boundary is accepted when the candidate-specific test does not reject the endpoint of the finite numerical search interval. A high boundary-acceptance rate therefore means that accepted components frequently reach the edge of the examined domain and are not localized inside it. It does not show that the population confidence region is globally unbounded, that every value outside the domain would be accepted, or that identification has failed. Conversely, if boundary acceptance disappears after the domain is widened, the accepted component is numerically localized within the wider range. Persistent boundary acceptance as successively wider finite domains are examined indicates continued lack of numerical localization over those investigated ranges.

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} [I_r(a_j) + I_r(a_{j+1})]. \quad (60)$$

The sum runs from $j = 1$ to 40 because the 41 grid points in G_0 define 40 adjacent grid intervals $[a_j, a_{j+1}]$. The trapezoidal sum includes separated accepted components without filling rejected gaps and does not interpolate critical-value crossings. It ranges from zero to four. The Monte Carlo mean of F_r is the frequency with which all tested grid points are accepted, not acceptance of the real line.

Coverage and power remain direct evaluations outside this grid construction. A post-main diagnostic examines sensitivity to the finite endpoints using

$$A_0 = [-1, 3], \quad A_1 = [-3, 5], \quad A_2 = [-5, 7]. \quad (61)$$

It uses $n = 1000$, 100 saved primitive samples, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and spacing 0.1. Stage 1 evaluates A_1 and its nested A_0 profile; Stage 2 reloads the same samples and adds only the tails needed for A_2 . Domain comparisons therefore change candidates, not samples. The outputs record accepted measure, grid share and count, boundary acceptance, all-grid acceptance, and incremental measure.

The boundary-acceptance diagnostic follows the finite-domain interpretation of Equations (58)–(59). The bounded diagnostic ends at A_2 , while main summaries retain A_0 and direct coverage and power remain unchanged.

5 Monte Carlo Results

5.1 Instrument Relevance and Point Estimation

Table 1 establishes that the four values of κ produce clearly ordered excluded-instrument relevance settings, using the first-stage F -statistic and partial R^2 as descriptive diagnostics.

Both diagnostics fall strongly as κ decreases. For $n = 500$, the median F -statistic falls from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$ and 1.22 at $\kappa = 0.10$; the corresponding end-points for $n = 1000$ are 212.82 and 2.04. Partial R^2 is comparatively stable across sample sizes, whereas F increases with n . The design therefore varies the instruments' explanatory contribution while sample size affects the amount of information available. These diagnostics describe relative relevance within the experiment; they are not quantile-specific weak-IV thresholds.

Table 2 is the central finite-sample performance table. It reports Bias, MAE, RMSE, Coverage, and the median accepted-set measure for $n = 1000$, $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and Oracle-GMM, Full-GMM, and DML-IVQR-BC.

As instrument relevance declines, MAE and RMSE generally increase for all three estimators and across the reported quantiles. At $\tau = 0.50$, moving from $\kappa = 1$ to $\kappa = 0.10$ raises RMSE from 0.072 to 0.945 for Oracle, from 0.100 to 1.018 for Full, and from 0.104 to 0.991 for DML. Oracle deterioration shows that the phenomenon is not generated solely by high-dimensional nuisance estimation: knowing the relevant controls does not replace instrument

Table 1. Instrument-strength calibration

n	κ	Median F	F : p10–p90	Median partial R^2	Partial R^2 : p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR-BC	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR-BC	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR-BC	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR-BC	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR-BC	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR-BC	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR-BC	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR-BC	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR-BC	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR-BC	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR-BC	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR-BC	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention ($\text{Bias} = \alpha_0 - \widehat{\alpha}$). Coverage is evaluated directly at the true structural quantile effect. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

information. Under the frozen convention $\text{Bias} = \alpha_0 - \widehat{\alpha}$, positive Bias means that estimates are below the truth on average.

5.2 Confidence-Set Informativeness and Power

Figure 2 shows the median grid-based accepted-set measure within $A_0 = [-1, 3]$. Panels (a) and (b) report $n = 500$ and $n = 1000$, respectively, allowing comparison across sample sizes for the same quantiles and estimators. The plotted quantity measures acceptance inside the four-unit primary domain, not unrestricted confidence-region length.

The measure rises strongly as κ falls, and at $\kappa = 0.10$ accepted sets often occupy nearly the whole primary domain. For $n = 1000$ and $\tau = 0.50$, the Oracle, Full, and DML medians rise respectively from 0.200, 0.400, and 0.300 at $\kappa = 1$ to 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Increasing n generally improves localization, but does not solve the weakest-relevance case. The accepted-set measure is deliberately defined on the fixed primary domain $A_0 = [-1, 3]$; the separate domain diagnostic is reserved for the Appendix.

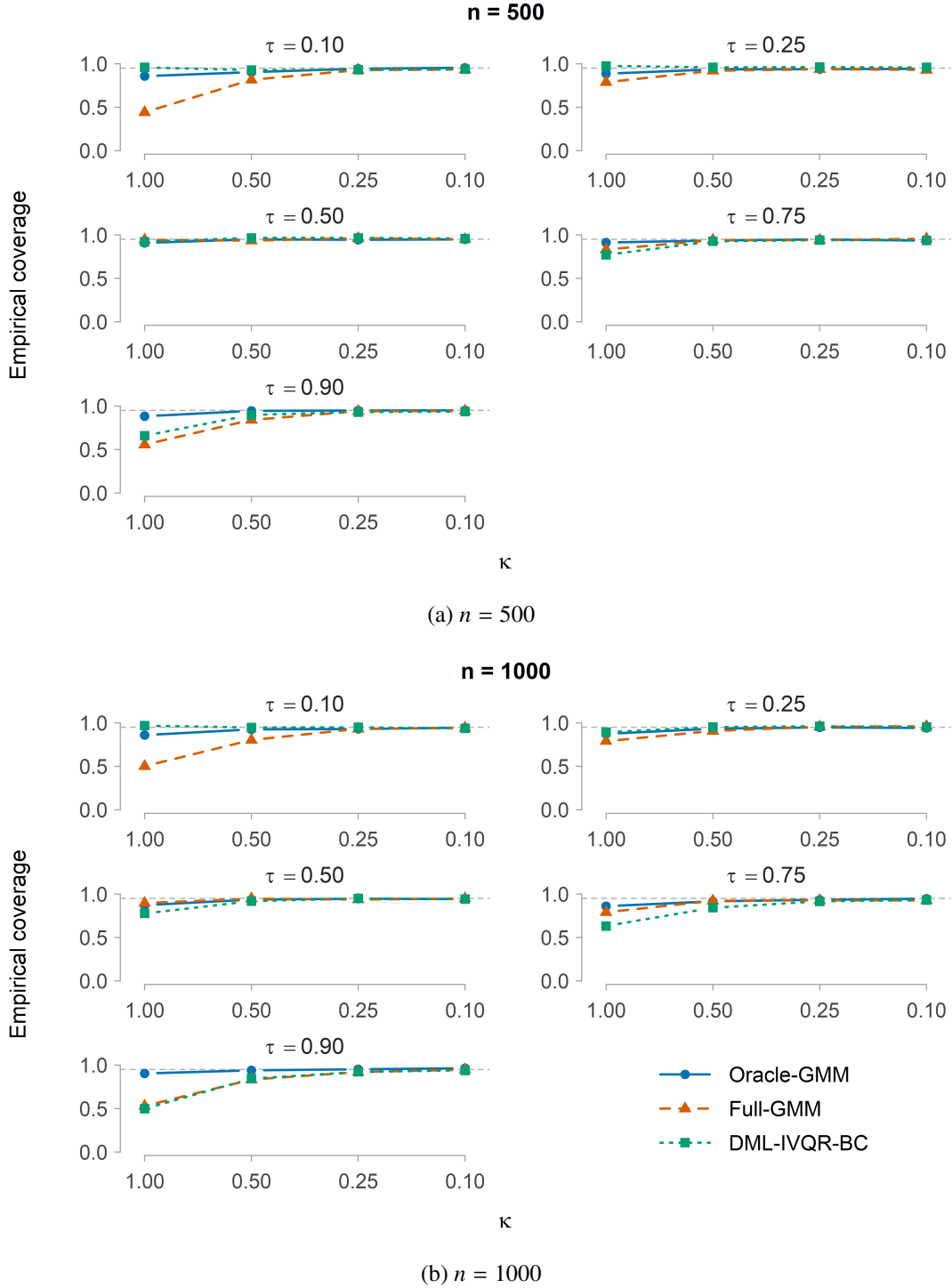
Figure 3 reports rejection probabilities under moderate false alternatives for $n = 1000$: panel (a) uses $\Delta = -0.50$ and panel (b) uses $\Delta = +0.50$. Rejection probabilities fall strongly as κ decreases, so false candidate treatment effects survive more frequently.

At $\tau = 0.50$ and $\Delta = -0.50$, power falls from 1.000 to 0.082 for Oracle, from 0.998 to 0.076 for Full, and from 1.000 to 0.104 for DML as κ falls from 1 to 0.10. At the positive alternative, the corresponding weakest-relevance probabilities are 0.092, 0.098, and 0.094. Positive and negative alternatives need not behave symmetrically because the candidate-specific moments can distinguish equally sized departures differently across directions. The accepted-set and power results are two complementary manifestations of declining informativeness.

5.3 Coverage and Finite-Sample Calibration

Figure 1 and the Coverage column of Table 2 show that empirical coverage is not uniformly close to the nominal level of 0.95. Coverage is evaluated directly at the true structural quantile effect, independently of the numerical grid. Weak-identification robustness of the asymptotic procedure does not imply exact finite-sample coverage at the simulated sample sizes.

Because each reported coverage rate is based on $R = 500$ Monte Carlo replications, the simulation itself introduces sampling uncertainty. For an estimated rejection or coverage prob-

**Figure 1.** Empirical coverage across instrument-strength designs.

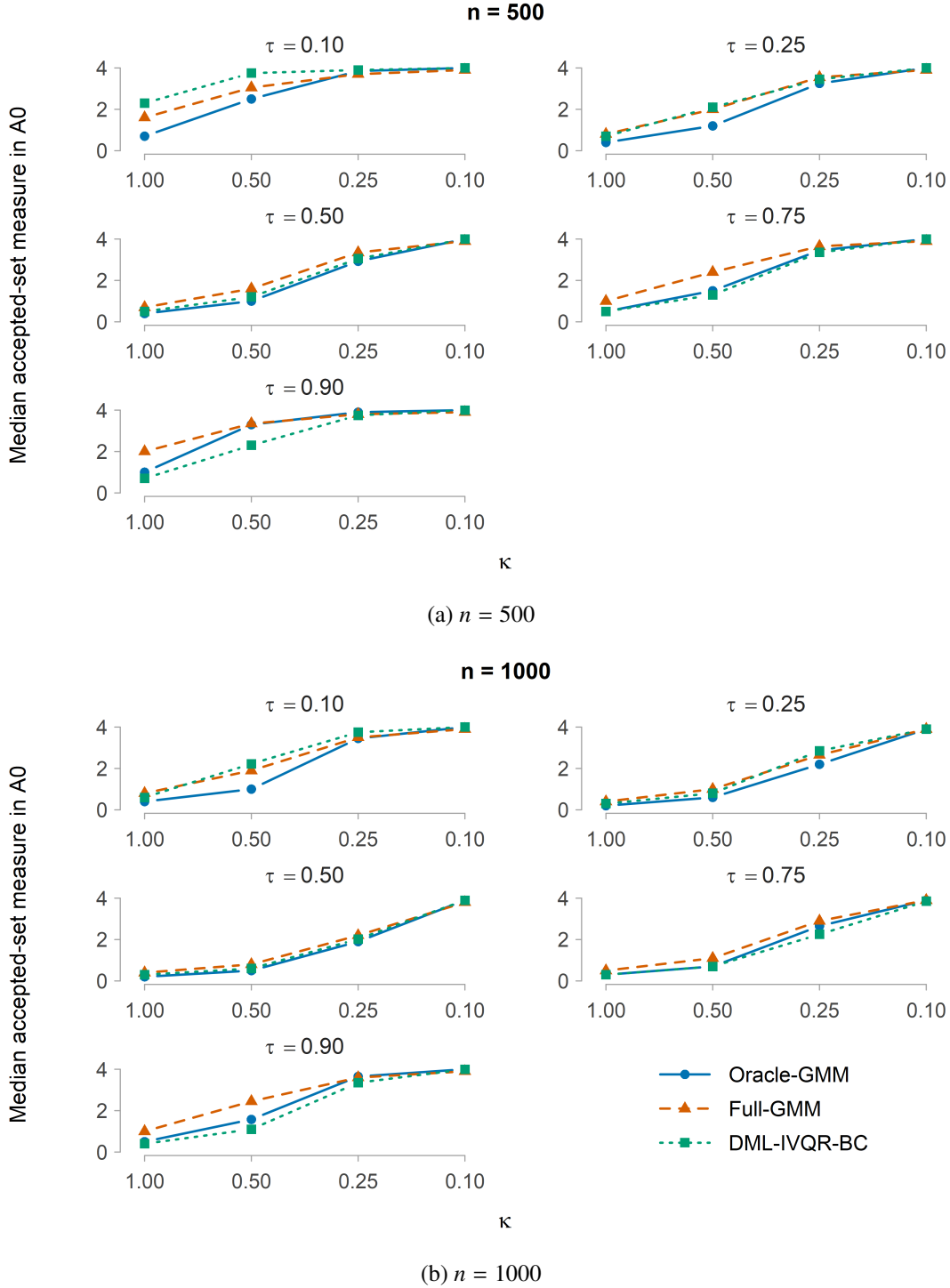


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

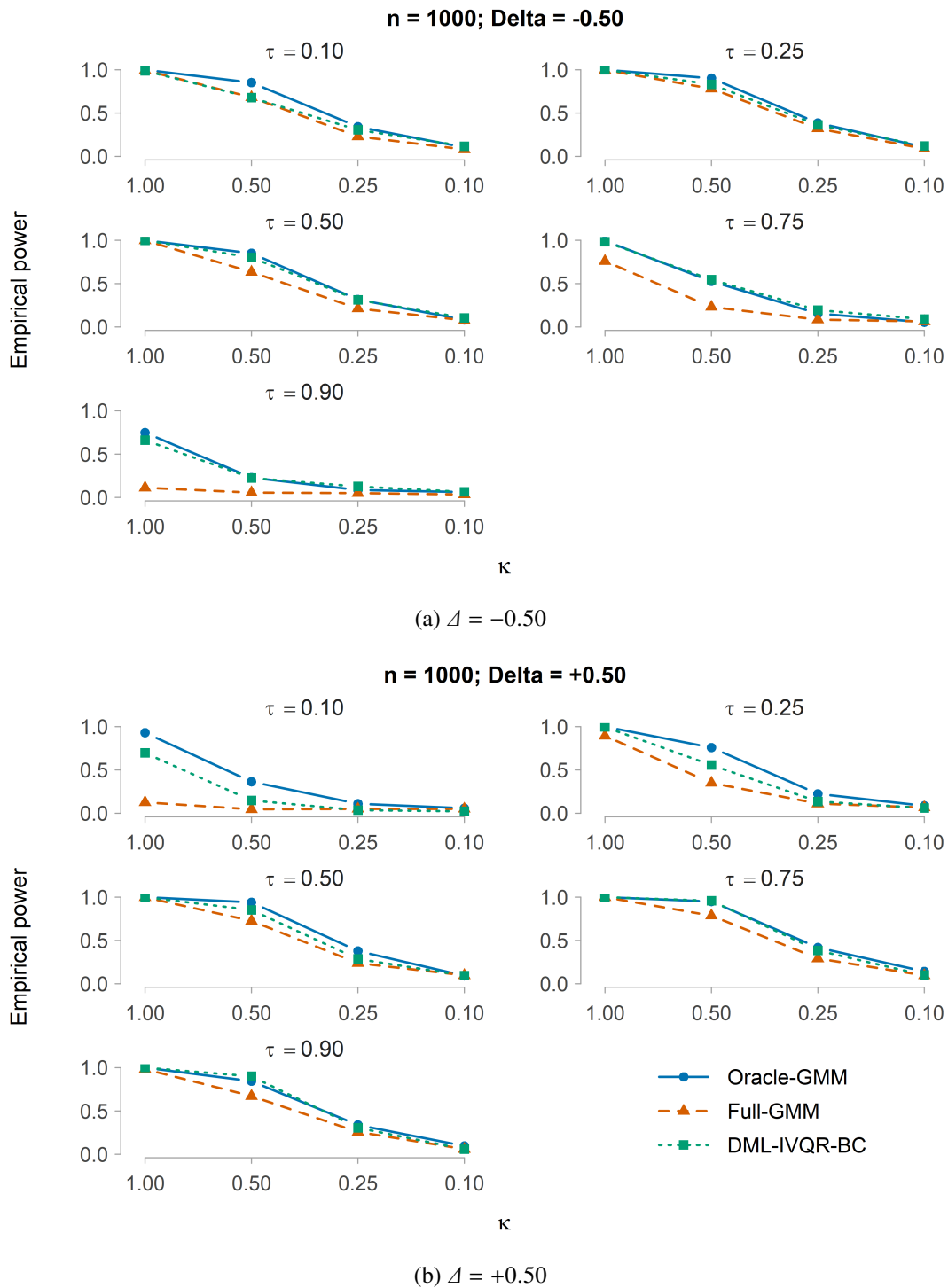


Figure 3. Rejection probability under moderate false alternatives, $n = 1000$.

ability $\widehat{p}$, the Monte Carlo standard error is

$$\text{MCSE}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1 - \widehat{p})}{R}}.$$

With $R = 500$, the largest possible MCSE is approximately 0.022, attained near $\widehat{p} = 0.5$; at the nominal coverage probability 0.95, the corresponding MCSE is approximately 0.010. Consequently, very small differences between nearby coverage rates should not be given substantive weight. The large deviations emphasized here are materially larger than Monte Carlo uncertainty: coverage values around 0.50, 0.63, or 0.86, for example, remain far from the nominal 0.95 benchmark relative to their simulation standard errors.

Distortions occur even at $\kappa = 1$ and vary substantially by estimator and quantile. For $n = 1000$, Full coverage is 0.502 at $\tau = 0.10$ and 0.530 at $\tau = 0.90$. DML coverage is 0.632 at $\tau = 0.75$ and 0.498 at $\tau = 0.90$, but 0.968 at $\tau = 0.10$. Oracle coverage is 0.860 at both $\tau = 0.10$ and $\tau = 0.75$, showing that high-dimensional nuisance estimation is not the sole explanation for undercoverage.

As relevance becomes very weak, the test may reject few candidates, retaining the truth frequently while accepted sets cover most of the primary domain. Such high coverage measures calibration and can coexist with very low informativeness and power.

5.4 Quantile Heterogeneity and Estimator Comparison

Taken together, Table 2 and Figures 1–3 show that weakening relevance worsens performance across quantiles, although the magnitude differs. Tail quantiles often display larger estimation errors or more severe calibration problems, but there is no universal ordering between the center and tails or between the two tails.

Oracle generally provides the strongest benchmark, yet deteriorates sharply as relevance weakens. DML often improves on Full, but does not dominate it in every cell or reproduce Oracle uniformly. These differences cannot be attributed solely to regularization because the estimators also use different residualization conventions.

Weaker excluded-instrument relevance raises point-estimation error, reduces the ability to reject false candidate values, and expands the set of accepted treatment-effect values. These effects occur for Oracle, Full, and DML-IVQR and vary across quantiles. Regularized nuisance estimation can improve performance relative to the unregularized high-dimensional benchmark in some cells, but it cannot replace missing instrument information.

6 Conclusion

This thesis studies how systematically weakening excluded-instrument relevance affects the finite-sample performance and informativeness of weak-identification-robust inference in high-dimensional IVQR. The Monte Carlo extension varies a single relevance parameter, κ , while preserving the marginal treatment distribution, the endogeneity channel, the structural outcome equation, the true quantile treatment effects, the controls, and the observed instrument construction of the baseline design in Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR-BC are evaluated on identical simulated samples across five quantiles and two sample sizes. This design isolates the consequences of progressively reducing the contribution of the excluded instruments to treatment without interpreting κ as a complete measure of quantile-specific IVQR identification.

The main result is clear. As excluded-instrument relevance declines, point estimation becomes substantially less accurate and candidate-specific inference becomes less informative. MAE and RMSE increase for all three estimators, including Oracle-GMM. At the median with $n = 1000$, for example, Oracle RMSE rises from 0.072 at $\kappa = 1$ to 0.945 at $\kappa = 0.10$, while DML-IVQR-BC rises from 0.104 to 0.991. The deterioration of the oracle benchmark shows that this pattern is not generated solely by high-dimensional nuisance estimation. Reduced instrument information directly weakens the ability of the IVQR moments to distinguish the true treatment effect from competing candidate values.

The inference results reinforce this conclusion. The grid-based accepted-set measure within the prespecified domain $A_0 = [-1, 3]$ expands sharply as relevance falls, while rejection probabilities against false treatment-effect values decline. At the weakest relevance setting, accepted values frequently occupy almost the entire primary domain and power against moderate false alternatives is low even at $n = 1000$. Larger samples improve localization and rejection power when usable instrument signal remains, but they do not eliminate the loss of informativeness at $\kappa = 0.10$. The magnitude of this deterioration also varies across quantiles: tail quantiles are often more difficult than the median, but there is no universal ordering across all estimators and relevance levels.

A separate finding concerns finite-sample calibration. Empirical coverage is not uniformly close to the nominal 0.95 level, including in some cells with the strongest relevance setting. Undercoverage also appears for Oracle-GMM, which rules out high-dimensional regularization as the sole explanation. With $R = 500$ replications, the maximum Monte Carlo standard error of a reported coverage probability is approximately 0.022, so the large coverage deviations documented in the simulations cannot be explained by Monte Carlo noise alone. These

findings indicate that the asymptotic chi-square reference distribution can be imperfectly calibrated at the simulated sample sizes. At the same time, very high coverage under weak relevance is not necessarily desirable: it can coexist with broad accepted sets and very low power against false candidates.

The estimator comparison clarifies both the usefulness and the limits of DML-IVQR. Regularized nuisance estimation often improves performance relative to unregularized Full-GMM, but DML-IVQR-BC neither dominates in every design cell nor uniformly reproduces the Oracle benchmark. Because the estimators also differ in their residualization implementation, the observed gaps cannot be interpreted as the isolated causal effect of regularization. More fundamentally, Neyman orthogonality protects the target moment against first-order nuisance-estimation error; it does not supply the identifying information lost when excluded instruments become weakly relevant.

The scope of these conclusions is deliberately limited. The analysis uses one preserved DGP, two sample sizes, one same-sample DML-IVQR implementation, a finite numerical candidate domain, and the authors' asymptotic chi-square inference procedure. It does not study alternative instrument structures, cross-fitted implementations, alternative critical-value procedures, or an empirical application. These are natural directions for further work rather than unresolved requirements of the present experiment.

The central implication is both practical and methodological. In high-dimensional IVQR, successful nuisance regularization and weak-identification-robust test construction do not by themselves guarantee precise or informative inference. When excluded instruments contain little information about treatment, estimation error can rise sharply, false structural effects may be difficult to reject, and finite-sample calibration can remain imperfect. Applied work should therefore evaluate instrument relevance, point-estimation accuracy, coverage, and inferential informativeness separately. No single measure provides sufficient evidence of reliable structural inference.

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